SecuGen

Unity 20 Bluetooth Fingerprint Reader User Guide

Introduction, How to Use, How to Connect

FCC STATEMENT

FCC Declaration of Conformity

This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions:
(1) This device may not cause harmful interference.
(2) This device and its accessories must accept any interference received, including interference that may cause undesired operation.

Note:

This equipment has been tested and found to comply with the limits for a class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures.

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Part 15.21 statement

Changes or modifications not expressly approved by the manufacturer (or party responsible) for compliance could void the user's authority to operate the equipment

This device complies with RF exposure requirements.

Notice

Product Handling

1. Users should not modify, disassemble or repair the equipment and components at will.
2. In the following situations, the surrounding environment may cause heat. This may shorten the battery life or damage the product or cause a fire.
    - Do not operate or store the product at too low or too high of a temperature.
3. Do not expose the product to excessive heat.

Battery Handling

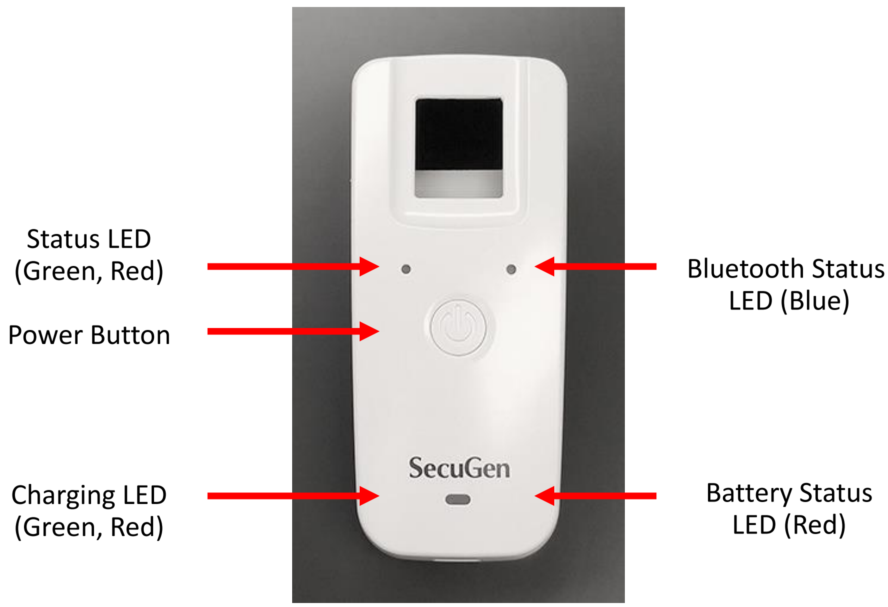
1. Charging the battery
    - Fully charge the battery when using for the first time.
    - To protect the built-in rechargeable battery, charge it at least once every 6 months. If too much time passes between charges, the life of the rechargeable battery may be reduced, or the rechargeable battery may no longer be able to be charged.
2. Do not use or leave the product at very high temperature conditions.
3. Battery life may be shorter in high or low temperature conditions.

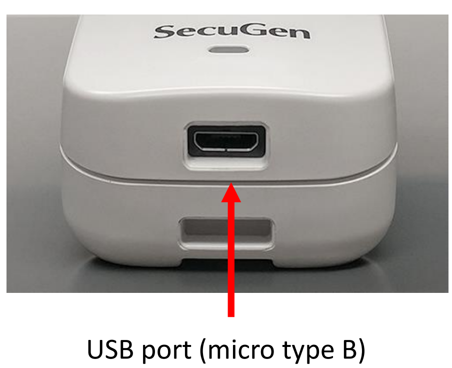
Table of Contents

1. Introduction

The Unity 20 Bluetooth is a stand-alone fingerprint recognition system that combines a high-performance optical fingerprint reader and Bluetooth module in one device. The Unity 20 Bluetooth can capture fingerprints and transfer fingerprint data to a host device such as a smartphone, tablet PC, iPhone or iPad via the Bluetooth interface.

The Unity 20 Bluetooth has a multi-functional power button, 4 LED indicators (2 indicators share one window), and a USB rechargeable battery as shown in the figures below.





The following tables describe the operation of the multi-functional power button and the LEDs.

Power Button Operation

Initial Status	Operation	Results
Powered Off	Long press - over 1.5 seconds	Power on Bluetooth Ready to Connect
Powered On	Short press	No response
Powered On / BT Ready	Long press	Power off
Powered On / BT Connected	Long press	Power off

Refer to Chapter 2 (How to Use the Bluetooth Reader) for more detailed instructions.

LED Indicators

LED Window	Color	Indication
Status LED	Green	Operation completed successfully
	Red	Operation failed
Bluetooth Status LED	Blue solid	Bluetooth connected
	Blue blinking **Bluetooth Classic mode**	Bluetooth is ready to connect **Even on-off blinking intervals**
	Blue blinking **BLE mode**	Bluetooth is ready to connect **Uneven on-off blinking intervals**
Charging LED	Red solid	Battery is charging
	Green solid	Battery is fully charged
Battery Status LED	Red blinking each second	Battery level is low

LED Indicators – When Powered Off

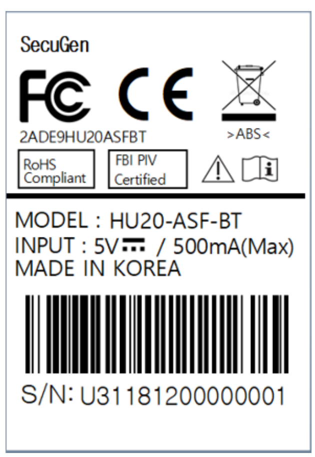
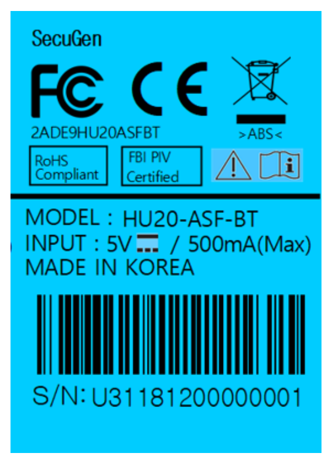
Device Status	Status LED	Bluetooth LED	Battery LED	Charging LED
USB unplugged	-	-	-	-
USB plugged in / charging	-	-	-	Red solid
USB plugged in / fully charged	-	-	-	Green solid

LED Indicators –When Powered On

Device Status		Status LED	Bluetooth LED	Battery LED	Charging LED
BT Ready	Unplugged, BAT OK	-	Blinking	-	-
	Unplugged, BAT LOW	-	Blinking	Blinking	-
	Plugged, fully charged	-	Blinking	-	Green solid
BT Connected	Unplugged, BAT OK	-	Solid	-	-
	Unplugged, BAT LOW	-	Solid	Blinking	-
	Plugged, fully charged	-	Solid	-	Green solid

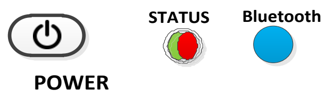
Label Information

Unity 20 Bluetooth supports two modes for data transfer: Bluetooth Classic mode and BLE mode. The two modes cannot be used at the same time. The label on the bottom of the product indicates the mode that the product is configured to support. When using a Bluetooth reader with your host device, the name of the Bluetooth reader will be displayed with the last 4 digits of its serial number as shown below.

Mode	Bluetooth Classic	BLE
Label		
Label color	White	Blue
Displayed name on host	Unity20BT-xxxx (Example: Unity20BT-0001)	Unity20BT-BLE-xxxx (Example: Unity20BT-BLE-0001)

2. How to Use the Bluetooth Reader

May 2019 version

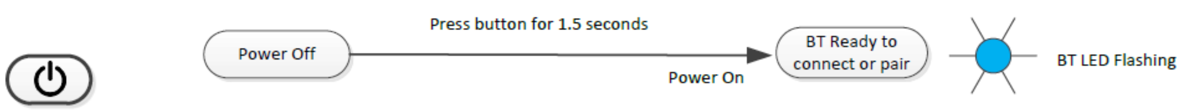


Power On

To power on the reader, press the power button for at least 1.5 seconds

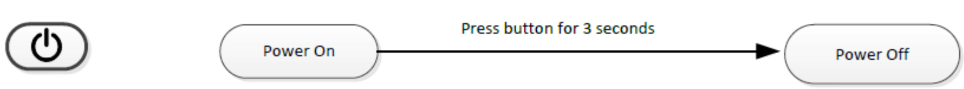
Pair and Connect

When the Bluetooth reader is powered on, the Bluetooth LED will be blinking (BT Ready). The reader is "Ready to Connect or Pair". If the reader has already been paired with a host device, use the connect function on your host device to complete the connection.



Power Off

To power off the reader, press the power button for 3 seconds

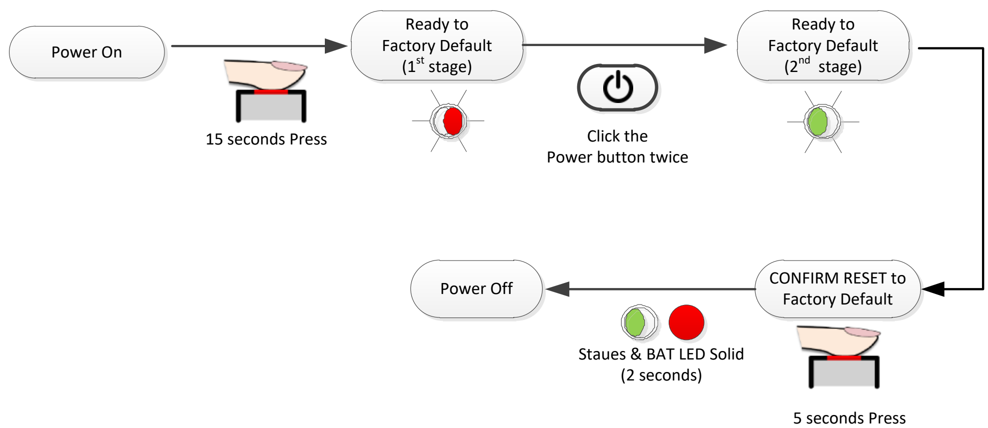


Factory Reset

To reset the Bluetooth reader and return to the original factory setting, do the following steps while the reader is on:

1. Place your finger on the sensor <u>for 15 seconds</u>. The Status LED will blink red. Remove your finger.
2. Click the Power button twice. The Status LED will change from red to yellow.
3. Place your finger on the sensor <u>again for another 5 seconds</u> to confirm the reset.

The Status LED and battery status LED will turn on solid for 2 seconds, and then the reader will power off indicating that the factory reset is completed.



If you want to cancel the factory reset process after the Status LED begins blinking (1st stage or 2nd stage), touch your finger on the sensor and the factory reset will be cancelled.

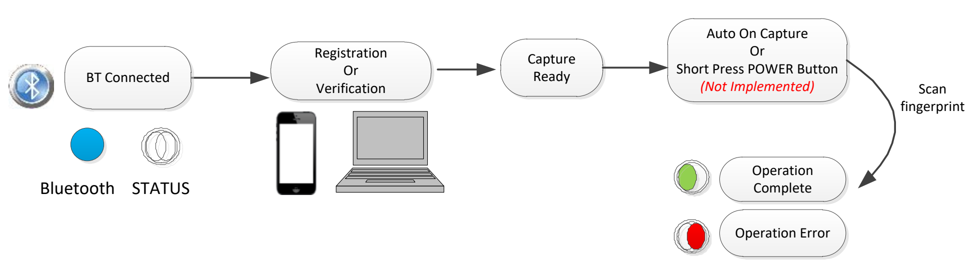
WARNING: Be careful when resetting the device. Performing these steps will clear all settings and delete all data from the Bluetooth reader.

Registration & Verification

After the Bluetooth reader is connected to a host device with appropriate software installed, you can register fingerprints or perform fingerprint recognition functions such as verification or authentication. (Refer to the next chapter for detailed instructions on how to connect to devices.)

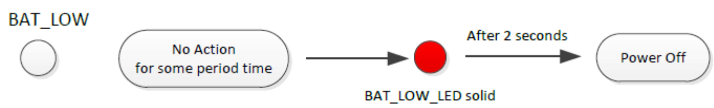
In general:

- If an operation is successfully completed, the Status LED will turn green.
- If an operation has failed, the Status LED will turn red.



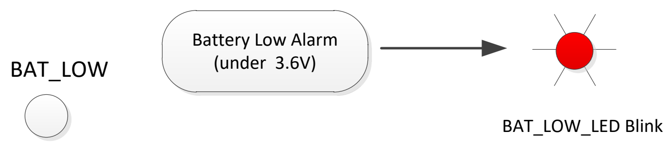
Auto Power Off

If there is no user activity for 10 minutes, the Battery Status LED will turn on solid red for 2 seconds, and then the device will automatically power off.



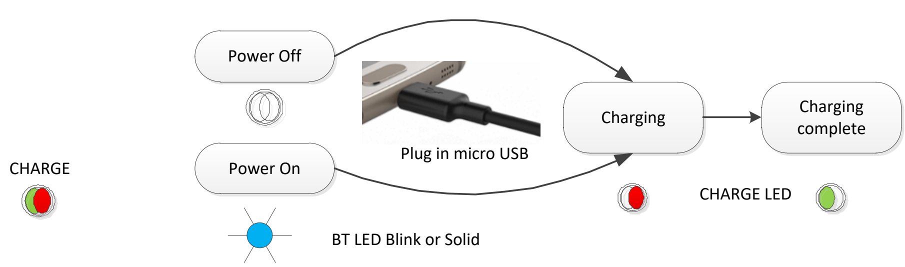
Battery Status

When the battery voltage drops below 3.6 V, the Battery Status LED will blink red once every second to indicate a low battery level. Connect to a USB charger to recharge the battery. To protect the battery, it is recommended to charge at least once every 6 months.



Charging Status

The Charging LED Status will turn on solid red while charging or solid green when the battery is fully charged. The Charging Status LED will stay on while the unit is plugged into a charger whether or not it is powered on.



3. How to Connect to Your Host Device

Unity 20 Bluetooth supports two modes for data transfer: Bluetooth Classic mode (SSP) and BLE mode (Bluetooth Low Energy). The two modes cannot be used at the same time. The following table shows which types of host devices are supported by each mode.

Host Device	Android	Apple iOS	PC
Bluetooth Classic mode	Yes	-	Yes
BLE mode	Yes	Yes	Yes

Click on the link to go directly to the instructions for how to connect.

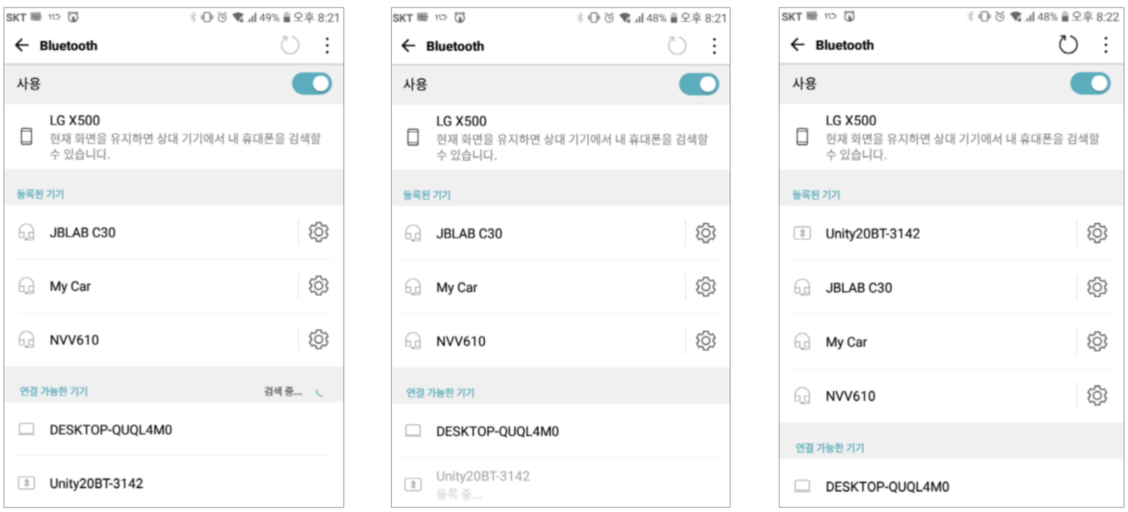
Bluetooth Classic mode – Connecting to an Android device

This section describes how to connect the Bluetooth reader to an Android device, while in Bluetooth Classic mode. These instructions are for SecuGen's Unity20 Bluetooth Demo app. If you are using a different app, follow the instructions provided by the app developer.

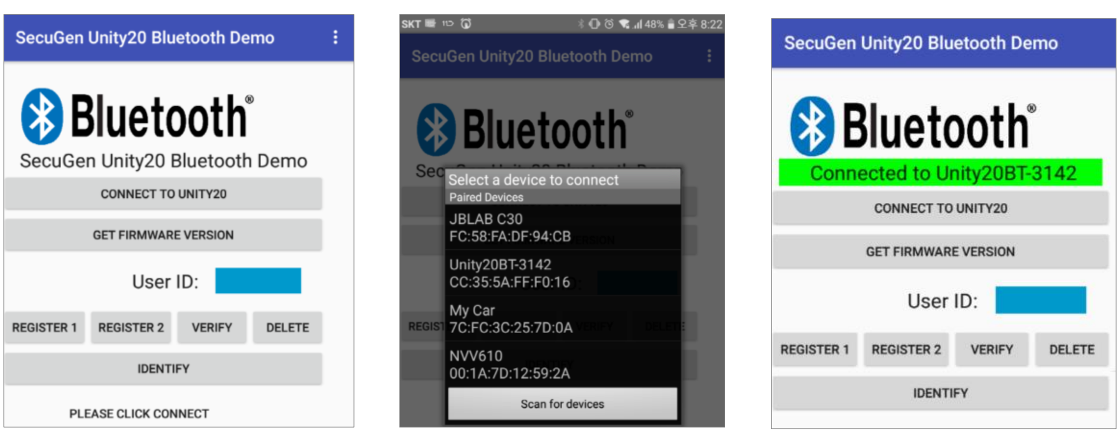
(1) Power on the Bluetooth reader. The Bluetooth Status LED should be blinking.
In Bluetooth Classic mode, the On-Off blinking time is even (0.5 second on, 0.5 second off).

(2) Turn on Bluetooth on the Android device.

(3) If you have not paired the Bluetooth reader with the Android device before, search for the reader from the Bluetooth menu of the Android device. When found, the Bluetooth reader name will be displayed as **Unity20BT-xxxx**, where xxxx is the last four digits of the serial number. In the example screens below, the device name is Unity20BT-3142. Click on the name of the Bluetooth reader to pair it.



(4) After pairing with the Bluetooth reader, connect it to the Android app. In the SecuGen Unity20 Bluetooth Demo app, click **CONNECT TO UNITY20**. A list of paired devices will be displayed. Click on the name of the Bluetooth reader you wish to connect. The screen will show **Connected to <name of your reader>** in green.

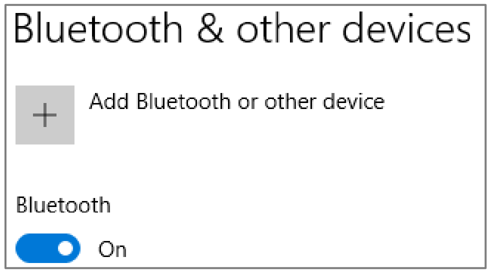


Bluetooth Classic mode – Connecting to a PC

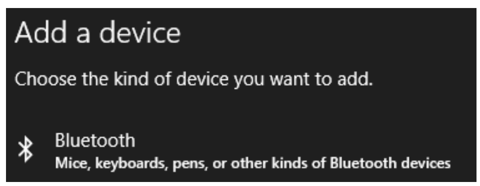
This section describes how to connect the Bluetooth reader to a PC with Bluetooth capability, while in Bluetooth Classic mode. These instructions are provided for Windows 10.

(1) Power on the Bluetooth reader. The Bluetooth Status LED should be blinking.
In Bluetooth Classic mode, the On-Off blinking time is even (0.5 second on, 0.5 second off).

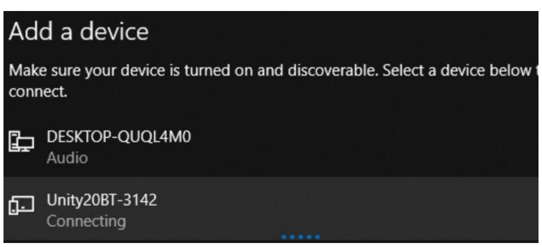
(2) Turn on Bluetooth capability on the PC.

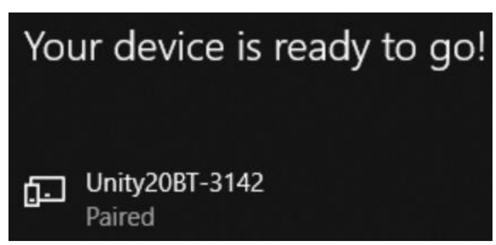


If you have not paired the Bluetooth reader with the PC before, search for the device on the PC. From the **Add a device** menu, select **Bluetooth**.



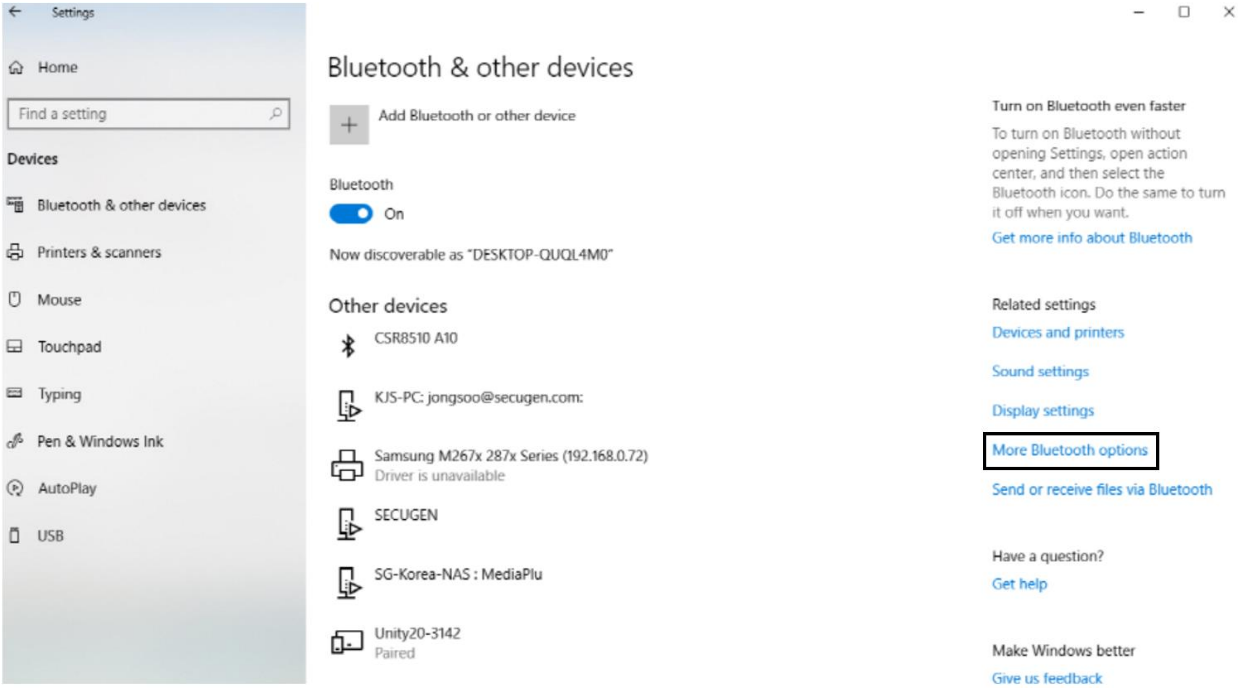
When found, the Bluetooth reader name will be displayed as **Unity20BT-xxxx**, where xxxx is the last four digits of the serial number. In the example screens below, the device name is Unity20BT-3142. Click on the name of the Bluetooth reader to pair it.



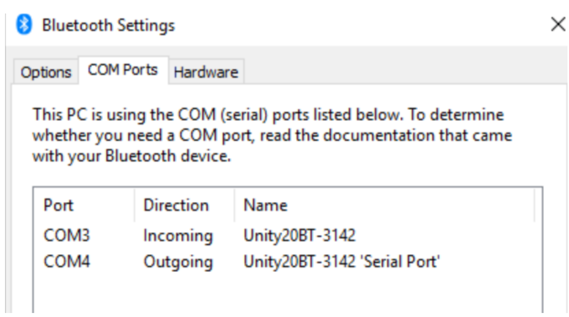


(3) Open the Bluetooth COM port.

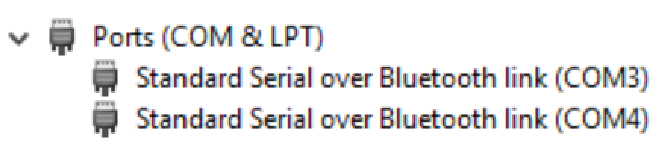
In the **Bluetooth & other devices** settings of the PC, select **More Bluetooth options**.



Select the **COM Ports** tab, and the assigned COM ports for the Bluetooth reader you have just paired will be displayed. Two serial ports are assigned to the Bluetooth reader–an outgoing serial port and an incoming serial port. The outgoing port is used by the PC to initiate the connection to the Bluetooth reader.



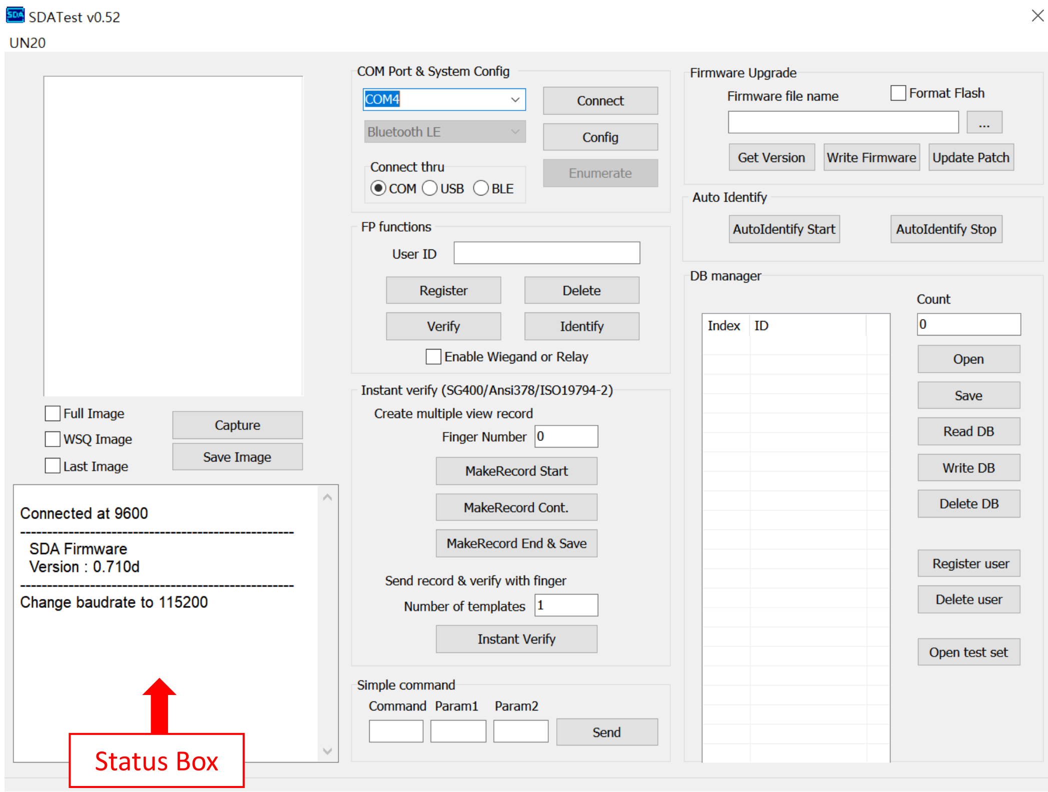
(4) After connecting the Bluetooth reader, you will see the newly assigned COM ports for Standard Serial Over Bluetooth in the Device Manager.



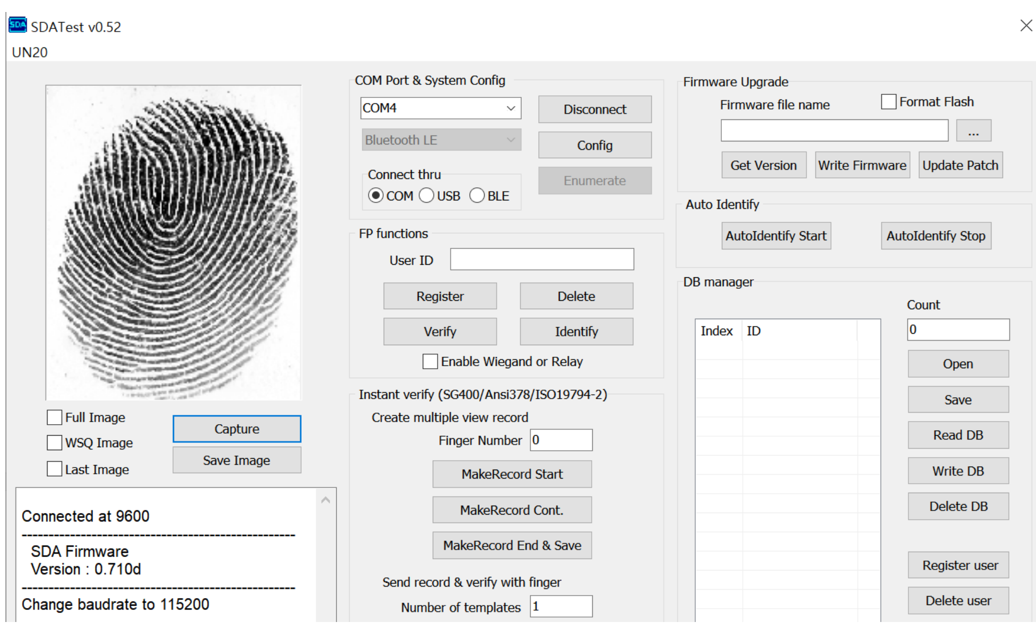
Since the Bluetooth reader uses the SPP (Serial Port Profile) Bluetooth profile for data transfer, the Bluetooth reader can wirelessly connect to the PC's COM port.

(5) Test Bluetooth communications by scanning a fingerprint.
- Run the latest version of the **SDATest.exe** utility program
- Choose the COM port assigned to the Bluetooth reader and click **Connect**

- When your PC has successfully connected to the reader via Standard Serial Over Bluetooth, the firmware version and connection baud rate will be displayed in the status box.



- Click **Capture**, and then place your finger on the sensor of the Bluetooth reader. The fingerprint image will be transferred to the PC from the Bluetooth reader via Bluetooth serial connection.



BLE mode – Connecting to an iOS device

This section describes how to connect the Bluetooth reader to an Apple iOS device, such as iPhone or iPad, while in BLE mode. These instructions are for SecuGen's Bluetooth Demo App. If you are using a different app, follow the instructions provided by the app developer.

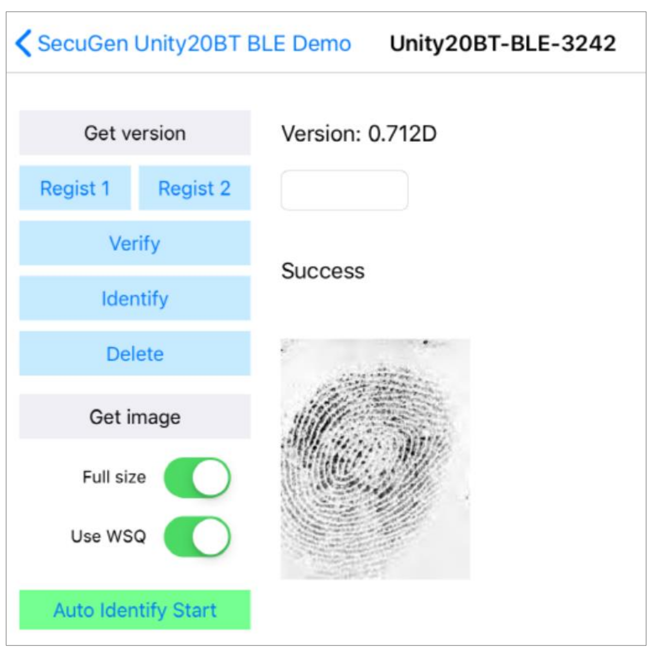
(1) Power on the Bluetooth reader. The Bluetooth Status LED should be blinking.
In BLE mode, the On-Off blinking time is uneven (2 quick blinks on, 1 long interval off).

(2) Turn on Bluetooth on the iOS device.

(3) Open the SecuGen Unity20BT BLE Demo app. The Bluetooth reader name will be displayed as **Unity20BT-BLE-xxxx**, where xxxx is the last four digits of the serial number. In the example screens below, the device name is Unity20BT-BLE-3242.

(4) Click on the Bluetooth reader name you wish to connect.

After connecting, the main screen of the demo app will appear. The demo app provides sample functions including image capture, register user, verify (1:1 matching) and identify (1:N matching).



BLE mode – Connecting to an Android device

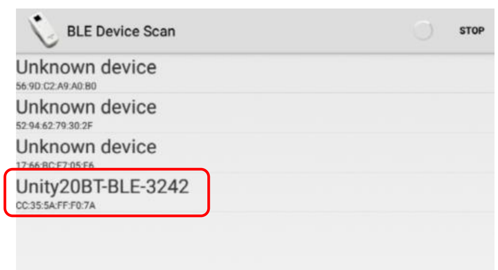
This section describes how to connect the Bluetooth reader to an Android device while in BLE mode. These instructions are for SecuGen's Bluetooth Demo App. If you are using a different app, follow the instructions provided by the app developer.

(1) Power on the Bluetooth reader. The Bluetooth Status LED should be blinking.
In BLE mode, the On-Off blinking time is uneven (2 quick blinks on, 1 long interval off).

(2) Turn on Bluetooth on the Android device.

(3) Open the SecuGen Unity20BT BLE Demo app. The Bluetooth reader name will be displayed as **Unity20BT-BLE-xxxx**, where xxxx is the last four digits of the serial number. In the example screens below, the device name is Unity20BT-BLE-3242.

(4) Click on the Bluetooth reader name you wish to connect.



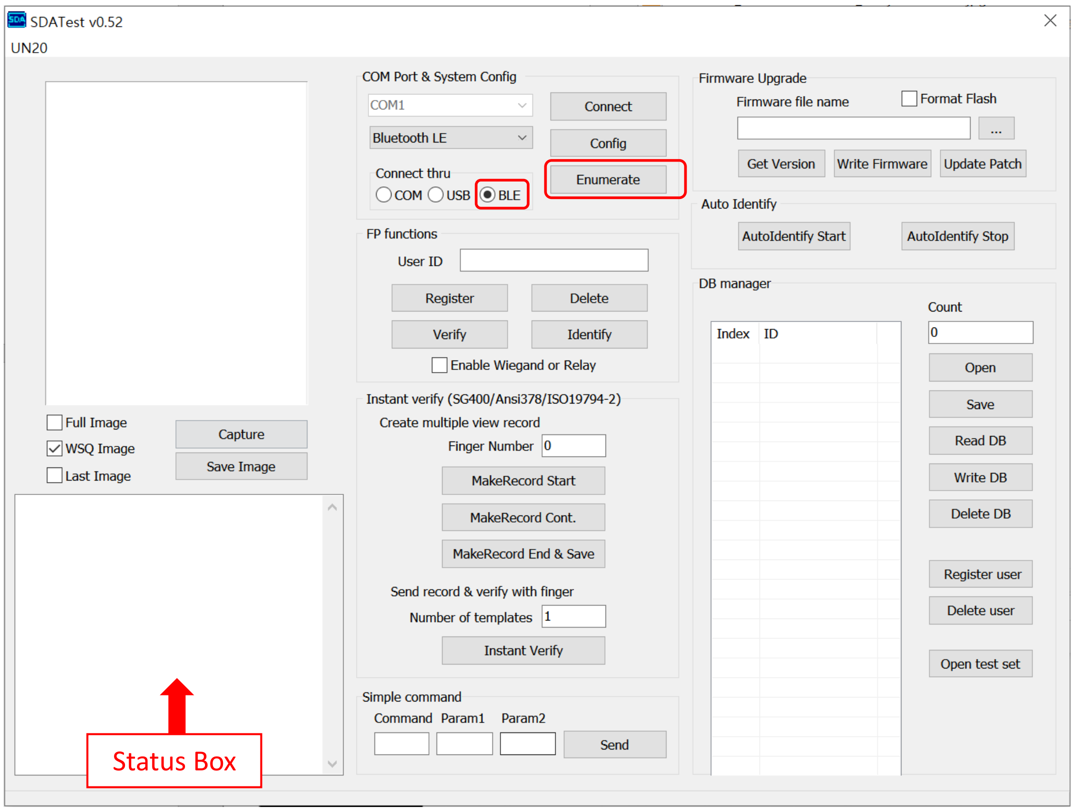
After connecting, the main screen of the demo app will appear. The demo app provides sample functions including image capture, register user, verify (1:1 matching) and identify (1:N matching).

BLE mode – Connecting to a PC

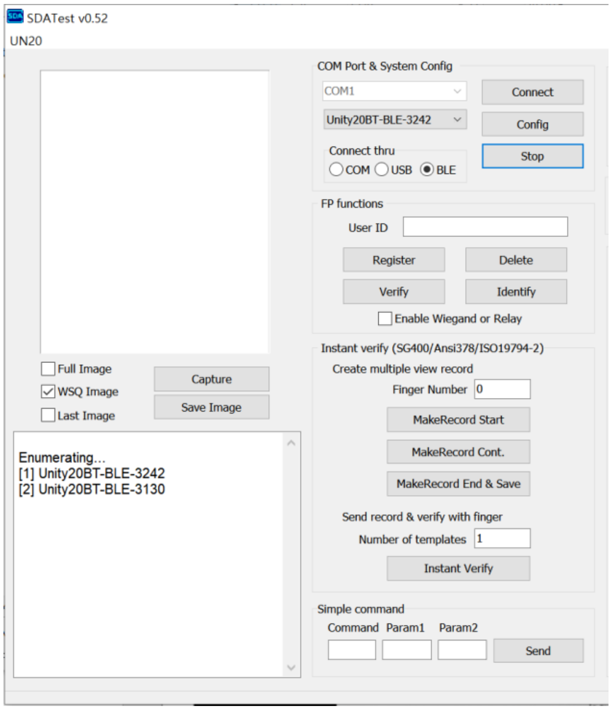
To connect the Bluetooth reader to a PC, while in BLE mode, requires using the SecuGen utility program, **SDATest.exe**.

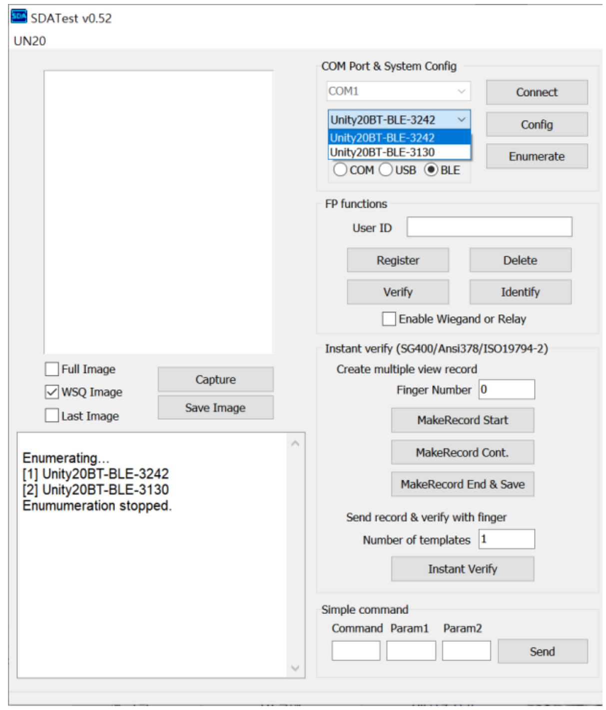
(1) Run the latest **SDATest.exe** utility program having the BLE connectivity function.

(2) Select **BLE** in the **Connect thru** menu, and then click **Enumerate**.

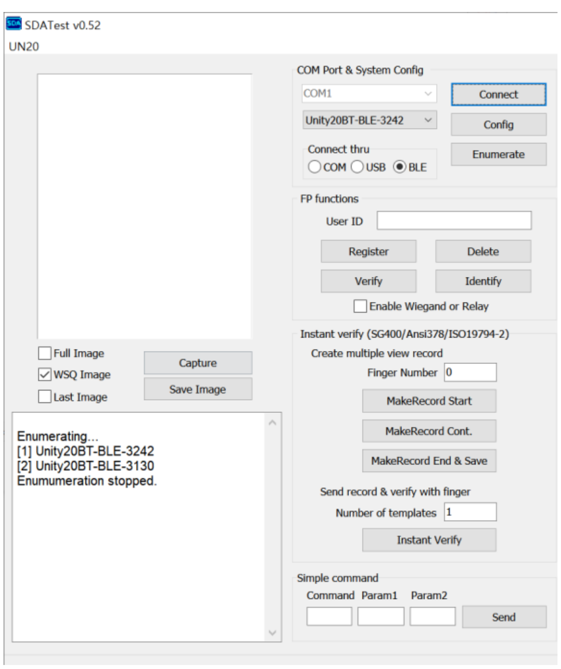


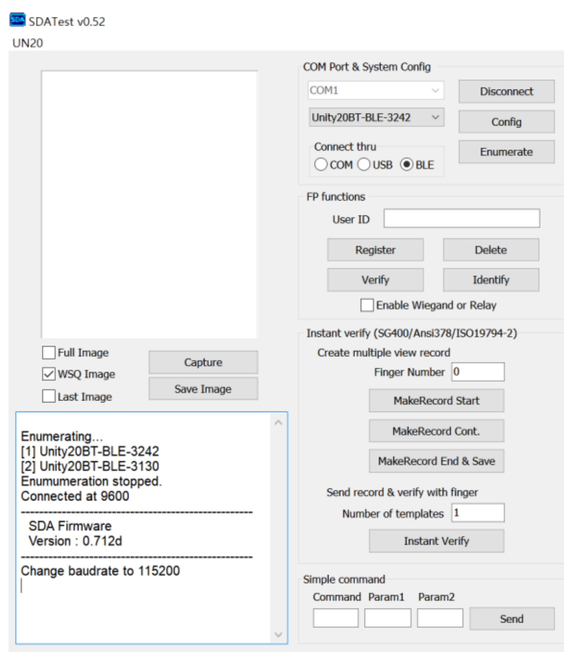
(3) Click on **Stop**. A list of Bluetooth reader(s) will be displayed in the status box. The Bluetooth reader name will be displayed as **Unity20BT-BLE-xxxx**, where xxxx is the last four digits of the serial number. In the example screens below, the device name is Unity20BT-BLE-3242.



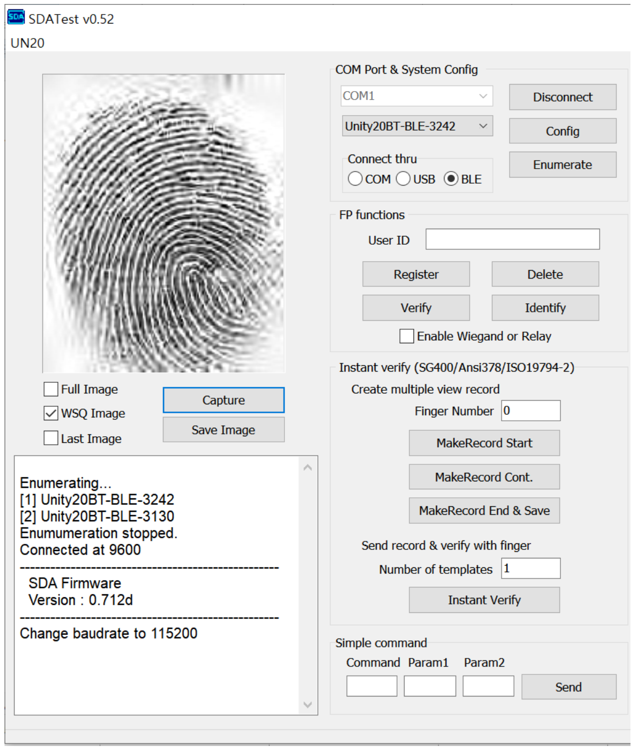


(4) From the drop-down list near the top, select the name of the Bluetooth reader you wish to connect, and click **Connect.** When your PC has successfully connected to the Bluetooth reader through BLE interface, the firmware version and connection baud rate will be displayed in the status box.





- Click **Capture**, and then place your finger on the sensor of the Bluetooth reader. The fingerprint image will be transferred to the PC from the Bluetooth reader via BLE interface.


SDATest v0.52
UN20
COM Port & System Config
COM1
Unity20BT-BLE-3242
Disconnect
Config
Enumerate
Connect thru
COM USB BLE
FP functions
User ID
Register
Delete
Verify
Identify
Enable Wiegand or Relay
Instant verify (SG400/Ansi378/ISO19794-2)
Create multiple view record
Finger Number 0
MakeRecord Start
MakeRecord Cont.
MakeRecord End & Save
Send record & verify with finger
Number of templates 1
Instant Verify
Simple command
Command Param1 Param2
Send
Full Image
WSQ Image
Last Image
Capture
Save Image
Enumerating...
[1] Unity20BT-BLE-3242
[2] Unity20BT-BLE-3130
Enumumeration stopped.
Connected at 9600
SDA Firmware
Version : 0.712d
Change baudrate to 115200